暨南大學
JINAN UNIVERSITY

Multimodal Language Technologies for Personalized Learning

From Knowledge Tracing to a Daily AI Study Companion

Prof. Zitao Liu

Dean, Guangdong Institute of Smart Education, Jinan University

2026 / 10 / 26

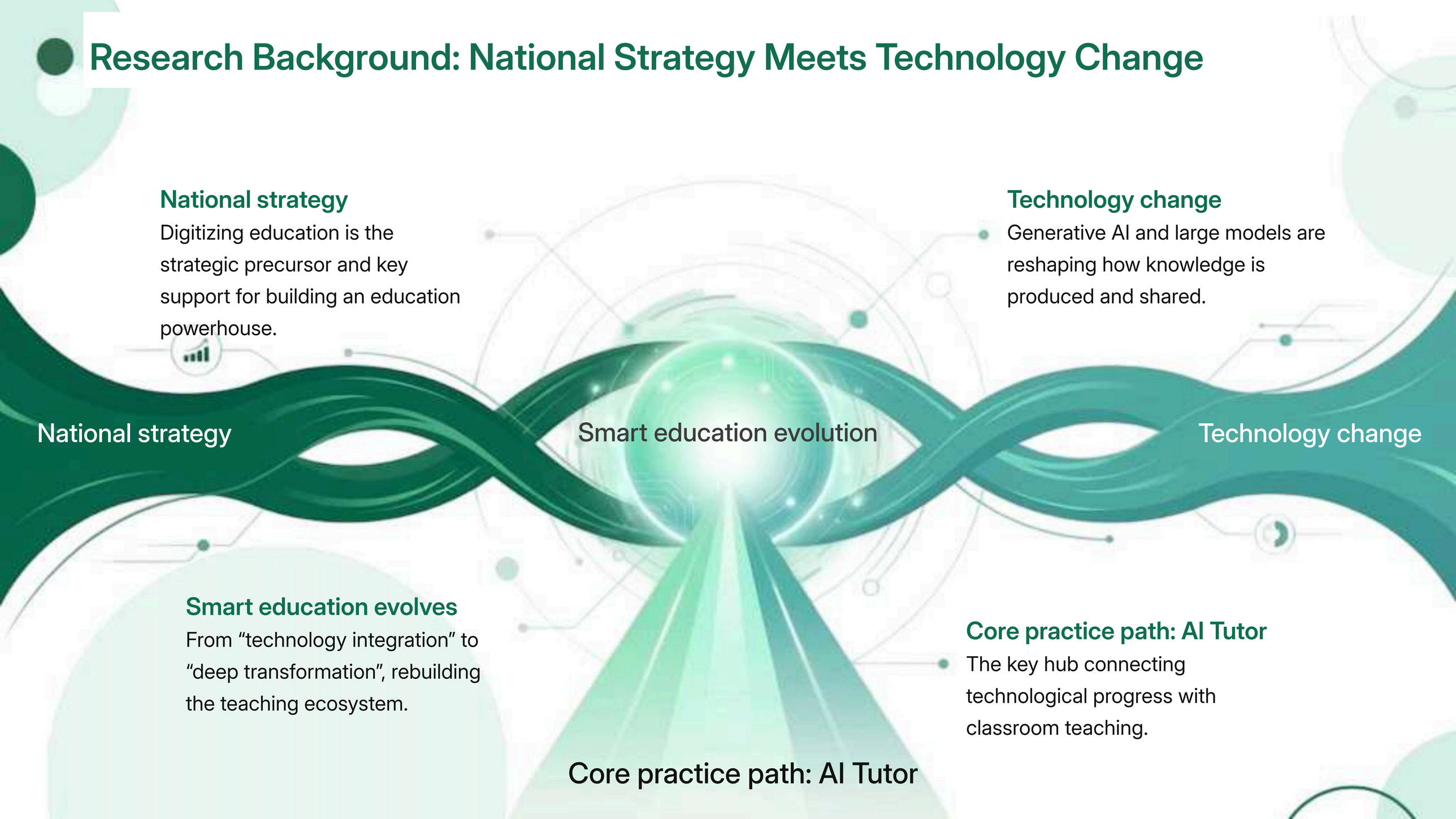
Research Background: National Strategy Meets Technology Change

National strategy

Digitizing education is the strategic precursor and key support for building an education powerhouse.

Technology change

Generative AI and large models are reshaping how knowledge is produced and shared.



Smart education evolves

From “technology integration” to “deep transformation”, rebuilding the teaching ecosystem.

Core practice path: AI Tutor

The key hub connecting technological progress with classroom teaching.

Guangdong Institute of Smart Education

Founded in May 2020 by the Guangdong Provincial Department of Education and Jinan University, the institute is a new type of research organization integrating research, talent development, technology development and achievement transfer. It now hosts the following provincial and ministerial key platforms:

- Guangdong-Hong Kong-Macao Joint Key Laboratory of Smart Education
- Guangdong Engineering Technology Research Center for Educational Big Data Services and Applications
- Greater Bay Area University Online Open Course Alliance

广 东 省 教 育 厅

广东省教育厅关于建立广东省智慧教育研究院的复函

暨南大学：

《关于申请建立广东省智慧教育研究院的函》收悉。经研究，我厅支持你校建立广东省智慧教育研究院；建议以省教育厅与暨南大学合作共建的模式建设广东省智慧教育研究院。

专此函复。

广东省教育厅

2020 年 5 月 19 日

（联系人：赵琦，联系电话：37628095）

暨 南 大 学 文 件

暨人〔2021〕95 号

暨南大学关于成立广东智慧教育研究院的通知

机关部处，直属单位，各校区、学部、学院、研究院：

经研究，学校决定与广东省教育厅共建成立广东智慧教育研究院。

特此通知

暨南大学

2021 年 6 月 18 日

Research Team

Leveraging the platform strengths of the Guangdong Provincial Department of Education and Jinan University, together with the research environment of the Greater Bay Area, we are building a research team dedicated to the reform of education informatization.

- Newly recruited talent is predominantly young and international.

Zitao Liu

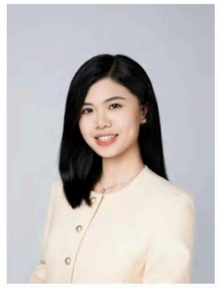
Quanlong Guan

Boyu Gao

Qianru Liang

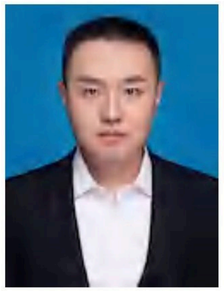
Zhiwei Yang

Teng Guo


Jiaqi Zheng

Tingjie Wan

Mingliang Hou


Wenzhou Dou

Agenda

01 LLM-driven Deep Knowledge Tracing and Personalized Learning Modeling

02 Multimodal Teaching Resource Generation with Educational Agents

03 Precision Learning Companion (OpenClaw) for Students

04 Summary and Outlook

01

LLM-driven Deep Knowledge Tracing and Personalized Learning Modeling

Core Tasks of Personalized Education

What do students know? → Knowledge Tracing

Modeling goal

Predict a dynamically changing probability of knowledge mastery.

Temporal nature

Emphasizes dynamic change, tracking how knowledge states evolve.

Core value

- Dynamic modeling: real-time mastery tracking for every knowledge point.
- Predicting future performance: forecasting answers on new items.

Who are the students? → Cognitive Diagnosis

Modeling goal

Portray a static, in-depth multi-dimensional cognitive profile.

Temporal nature

Focuses on static diagnosis, dissecting the structure of ability.

Core value

- Multi-dimensional profile: knowledge, behaviour and cognition combined.
- Fine-grained diagnosis: mastery change down to [illegible]

One dynamic, one static — complementary, together forming the technical foundation of personalized learning modeling

Research Framework

应用与反馈层

核心建模层

输入与建模层

真实教育场景

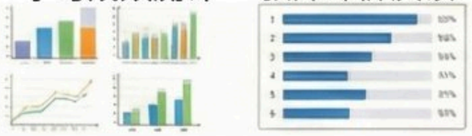
课堂使用　课中讲解　课后答疑

将模型能力应用于学习全流程，包括课前预习、课中答疑和课后诊断。

产出

效果评估

学习成效测评　教师评价反馈



通过学习成效测评和教师评价反进，系统地评估应用效果

指导

试点推广

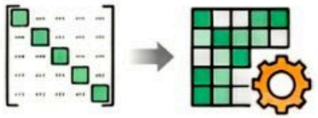
进行相应的系统研发与测试，在兴宁县10余所学校进行试点应用。

能力输出

反馈驱动，迭代优化

科学问题1：数据稀疏

研究工作一 (冷启动快速建模)

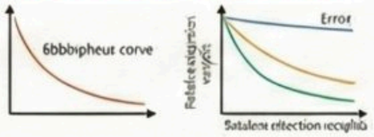


使用相对位置编码和锥注意力在冷启动场景下稳定预测

基础

科学问题2：同质化建模

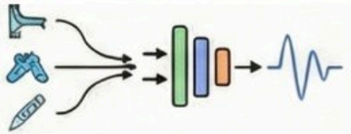
研究工作二 (个性化遗忘)



结合多种遗忘曲线来实现相对注意力机制

互补

研究工作三 (异质行为预测)

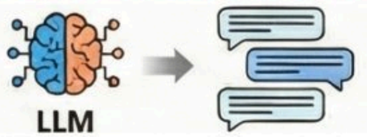


利用频率特征提取和异构专家网络实现个性化预测

增强

科学问题3：语义黑箱

研究工作四 (语义可解释诊断)



LLM

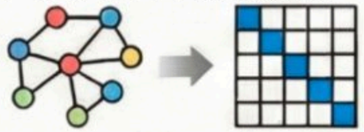
利用大语言模型进义语义对齐，生成诊断报告和学习建议

结构先验

行为建模

画像融合

知识结构数据



构建知识图谱与Q矩阵，提供知识点关联的结构先验，支撑冷启动场景下的快速建模。

融合

学习行为数据

采集作答序列、答题时间、练习频率等动态行为数据，驱动个性化遗忘与多模式行为建模。

融合

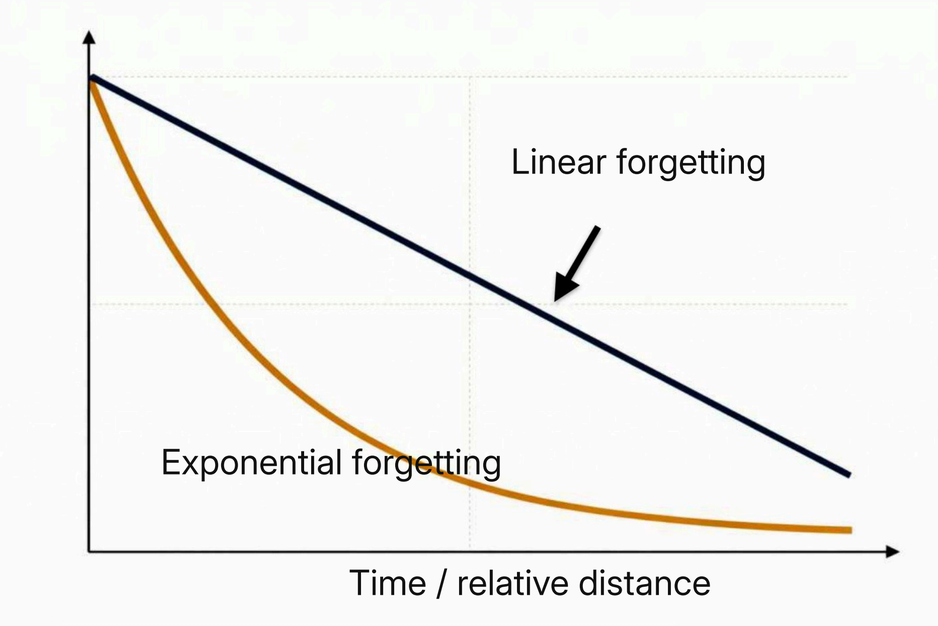
语义文本数据

构建知识、行为、认知的多维学习者画像，支撑LLM语义对齐与可解释诊断生成。

Research: A Knowledge Tracing Model with Relative Forgetting Attention

Research background & challenges

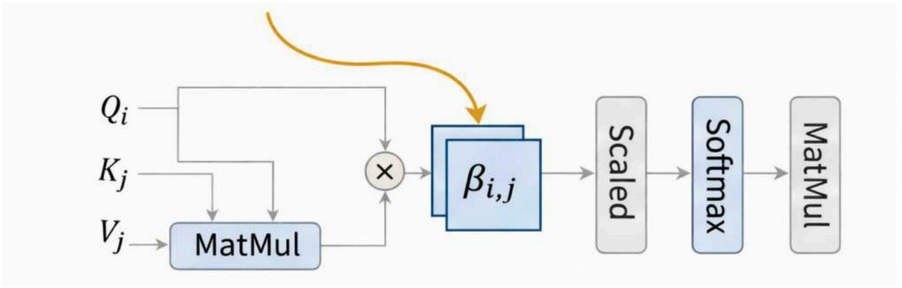
Forgetting: knowledge decays over time



Psychological research shows that student forgetting follows multiple patterns shaped by learning content and habit — complex knowledge tends to decay exponentially, while simple knowledge may decay linearly.

Limits of forgetting modelling in attention-based knowledge tracing

- Forgetting is coupled with content relevance — attention weights must encode both knowledge relevance and time-based forgetting, leaving responsibilities unclear.

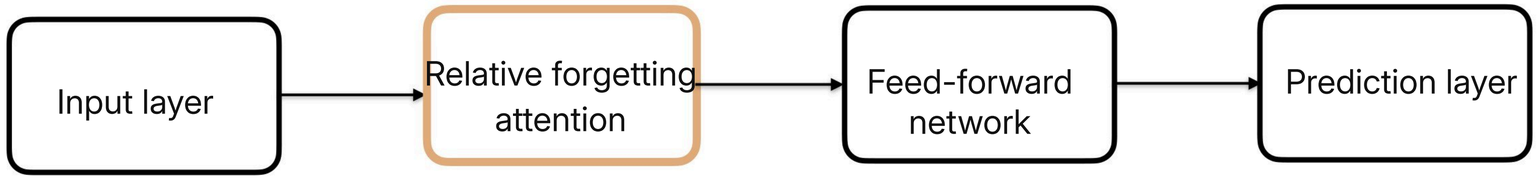


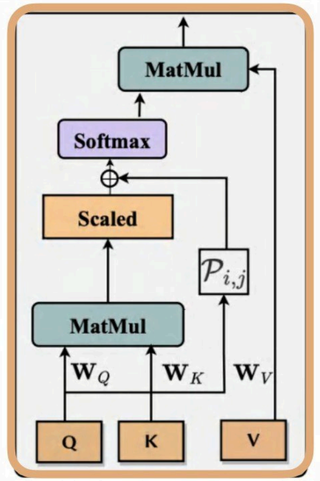
Monotonic attention (AKT)

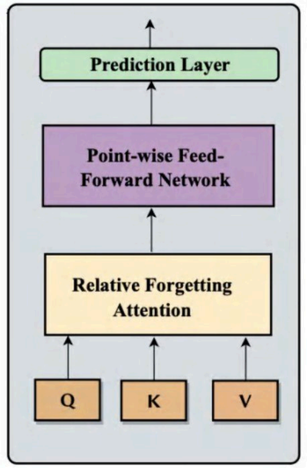
- Training on fixed-length windows, while student sequences keep growing — long-term changes in forgetting are hard to track.

Research: A Knowledge Tracing Model with Relative Forgetting Attention

Research idea: Decoupling “content” and “time” through Relative Forgetting Attention (RFA)







Decoupling

Forgetting is modelled separately from question relevance.

Flexibility

Adapts to multiple forgetting patterns, both linear and exponential.

Extrapolation

Strong length extrapolation for long sequences.

Contributions:

- General framework: a universal framework that enhances forgetting modelling for any attention-based knowledge tracing model.
- Personalized modelling: the decoupled design precisely captures individual and diverse forgetting behaviour.
- Long-term generalization: solves long-sequence generalization, making the model fit real long-term learning scenarios.

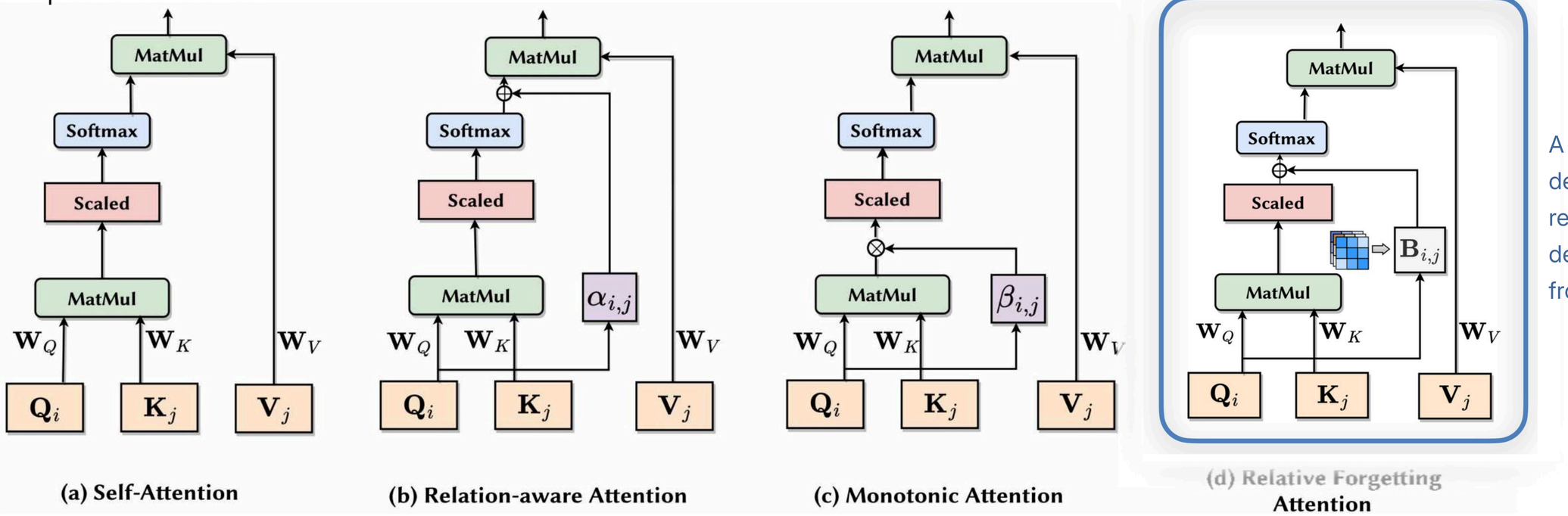
RFA separates forgetting modelling from attention-weight computation, so each component does its own job — enabling more precise and flexible personalized tracking.

Research: A Knowledge Tracing Model with Relative Forgetting Attention

Core mechanism: Relative Forgetting Attention (RFA)

$$\mathrm{RFA}(Q, K) = \mathrm{Softmax}\left(QK^{\intercal} / \sqrt{d_k} + B \right)$$

- RFA injects a flexible time-decay bias — implemented through Relative Position Encodings (RPEs) — directly into the attention scores, decoupling forgetting modelling from question relevance.



A decay bias that depends only on relative distance — decoupling forgetting from content

(a) Self-Attention: no notion of time or forgetting at all.

(b) & (c) RKT/AKT attention: time decay is multiplied into the Key or Value, coupling forgetting with content.

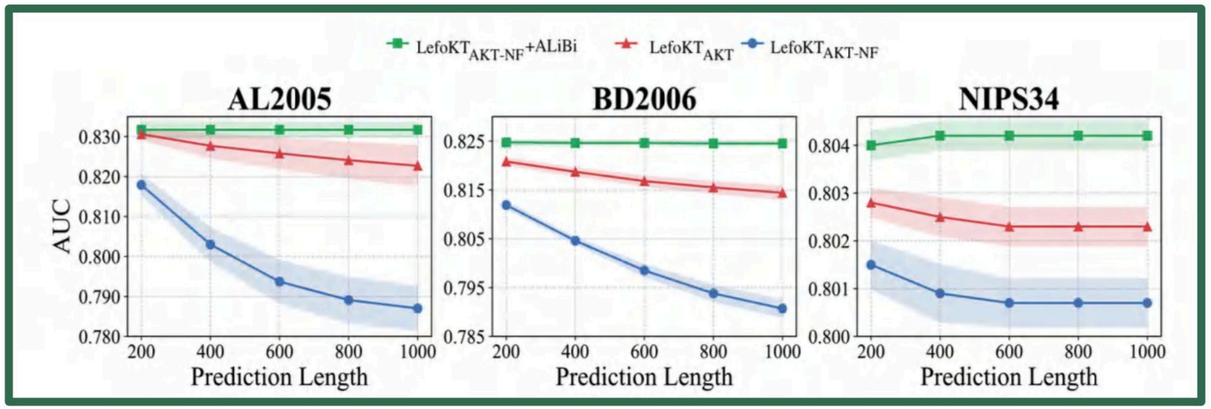
(d) Relative Forgetting Attention (RFA): the relative-distance decay bias B is added directly to the attention scores, decoupling forgetting from content.

Research: A Knowledge Tracing Model with Relative Forgetting Attention

Experimental results

1. Position encodings with multiple decay forms successfully simulate forgetting — RFA captures diverse forgetting patterns consistent with psychology.

Model	AL2005		BD2006		NIPS34	
	AUC	ACC	AUC	ACC	AUC	ACC
LefoKT$_{RKT}$	0.7761±0.0010	0.7964±0.0021	0.7631±0.0013	0.8468±0.0005	0.7966±0.0011	0.7264±0.0010
LefoKT$_{RKT-NF}$+ALiBi	0.7716±0.0008	0.7914±0.0007	0.7567±0.0045	0.8416±0.0046	0.7883±0.0006	0.7188±0.0007
LefoKT$_{RKT-NF}$+KERPLE-Log	0.7856±0.0009	0.8010±0.0008	0.7736±0.0019	0.8467±0.0005	0.7999±0.0002	0.7296±0.0002
LefoKT$_{RKT-NF}$+KERPLE-Power	**0.7857±0.0005**	**0.8013±0.0008**	**0.7740±0.0019**	0.8466±0.0004	**0.8003±0.0004**	**0.7297±0.0009**
LefoKT$_{RKT-NF}$+T5	0.7753±0.0008	0.7962±0.0005	0.7587±0.0026	0.8428±0.0025	0.7965±0.0005	0.7266±0.0006
LefoKT$_{RKT-NF}$+FIRE	0.7835±0.0011	0.8005±0.0009	0.7739±0.0010	**0.8475±0.0005**	0.8000±0.0001	0.7295±0.0007
LefoKT$_{RKT-NF}$+SandWich	0.7829±0.0014	0.8000±0.0015	0.7710±0.0010	0.8460±0.0012	0.7985±0.0004	0.7284±0.0004
LefoKT$_{AKT}$	0.8306±0.0019	**0.8124±0.0011**	0.8208±0.0007	0.8587±0.0005	0.8033±0.0003	0.7323±0.0005
LefoKT$_{AKT-NF}$+ALiBi	**0.8317±0.0021**	0.8110±0.0009	**0.8247±0.0006**	**0.8605±0.0012**	**0.8045±0.0003**	**0.7340±0.0004**
LefoKT$_{AKT-NF}$+KERPLE-Log	0.8276±0.0012	0.8115±0.0015	0.8230±0.0016	0.8603±0.0010	0.8040±0.0010	0.7333±0.0009
LefoKT$_{AKT-NF}$+KERPLE-Power	0.8258±0.0022	0.8094±0.0018	0.8207±0.0013	0.8588±0.0011	0.8041±0.0011	0.7338±0.0009
LefoKT$_{AKT-NF}$+T5	0.8260±0.0029	0.8073±0.0019	0.8201±0.0017	0.8589±0.0009	0.8035±0.0003	0.7326±0.0005
LefoKT$_{AKT-NF}$+FIRE	0.8221±0.0008	0.8097±0.0009	0.8116±0.0007	0.8559±0.0005	0.8027±0.0005	0.7319±0.0007
LefoKT$_{AKT-NF}$+SandWich	0.8275±0.0024	0.8100±0.0020	0.8234±0.0019	0.8600±0.0012	0.8037±0.0009	0.7334±0.0008



2. Our RPE is also useful for forgetting-based models such as AKT and RKT — replacing the original forgetting mechanism improves performance.

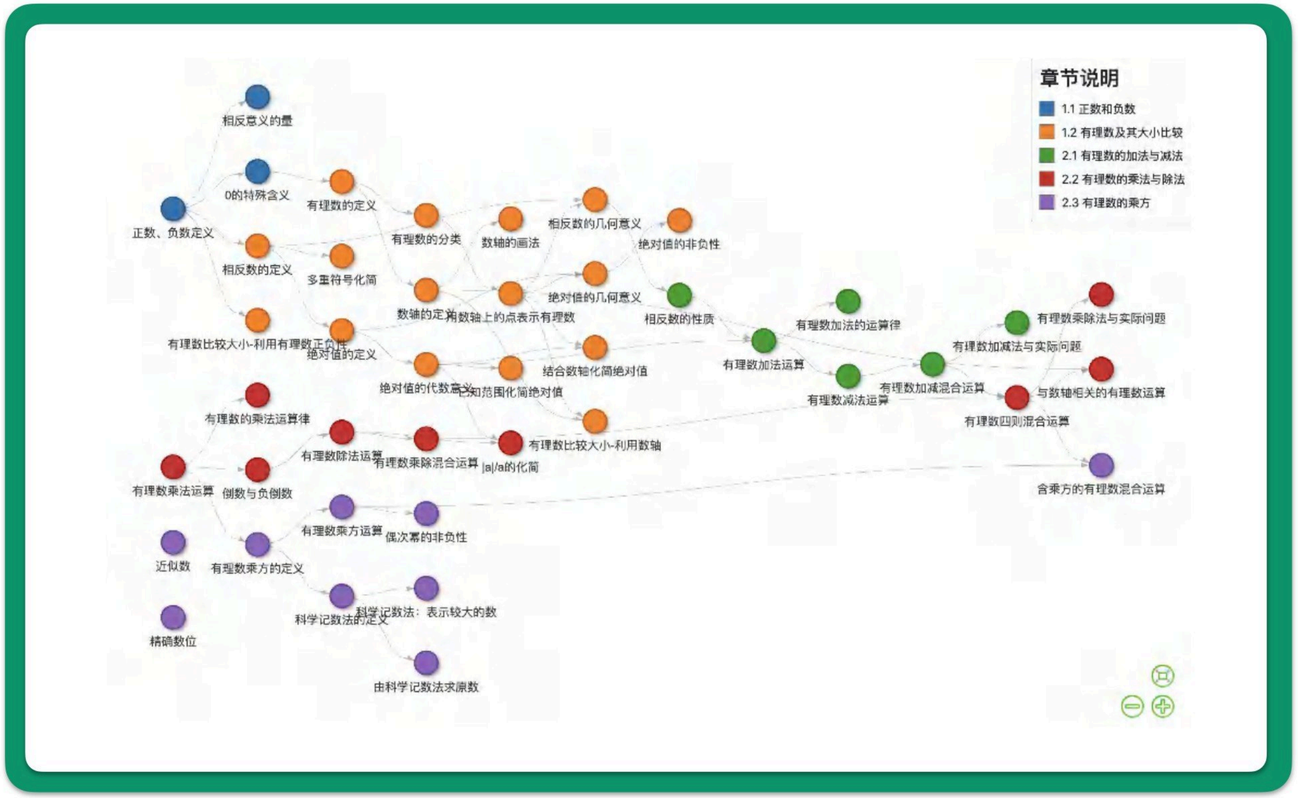
Model	AL2005	BD2006	NIPS34
LefoKT$_{RKT}$	0.7761±0.0010	0.7631±0.0013	0.7966±0.0011
LefoKT$_{RKT-NF}$+K-P	**0.7857±0.0005**	**0.7740±0.0019**	**0.8003±0.0004**
LefoKT$_{RKT}$+K-P	0.7837±0.0008	0.7735±0.0014	0.8000±0.0002
LefoKT$_{AKT}$	0.8306±0.0019	0.8208±0.0007	0.8033±0.0003
LefoKT$_{AKT-NF}$+ALiBi	**0.8317±0.0021**	**0.8247±0.0006**	**0.8045±0.0003**
LefoKT$_{AKT}$+ALiBi	0.8310±0.0010	0.8199±0.0008	0.8032±0.0002

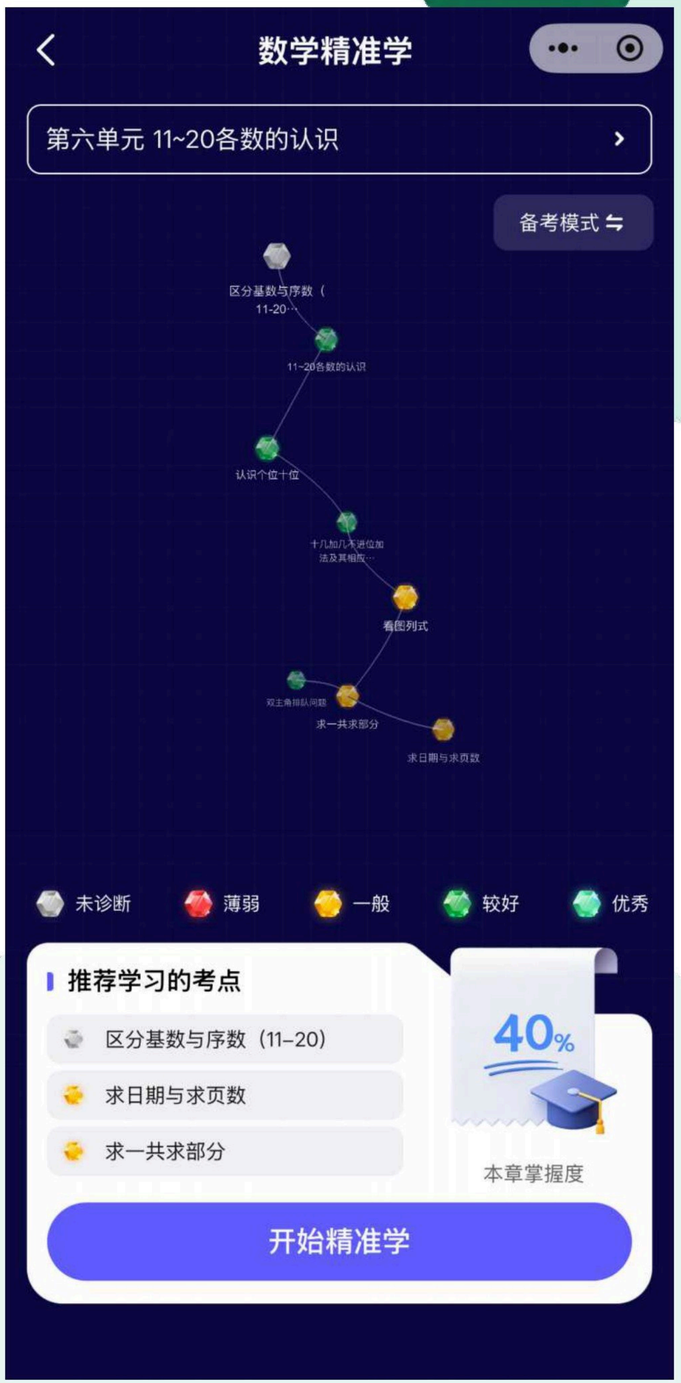
3. Ablation study: across sequence lengths, removing the corresponding module always degrades performance; the model with RPE predicts most stably, achieving effective extrapolation on long sequences.

Youheng Bai, Xueyi Li, Zitao Liu, Yaying Huang, Mi Tian, and Weiqi Luo. “Rethinking and improving student learning and forgetting processes for attention based knowledge tracing models.” AAAI 2025.

AI Precision Practice

Based on the large model's ability to tag knowledge points in questions and solution steps, we mine the relations between questions and knowledge tags, and identify the relations among all leaf knowledge points as the key support for precision practice.





Large-scale Pilot School Collaboration across Guangdong

兴宁市教师发展中心

关于举办人工智能赋能基础教育试点学校签约仪式暨专题培训的通知

各人工智能赋能基础教育试点学校：

为深入贯彻落实国家教育数字化战略行动，充分发挥人工智

能在推动基础教

智能素养和教育

设工作，经研究

培训。现将有关

一、培训目

通过系统培

模型开展教学创

学生诊断与精准

成果，打造兴宁

二、培训对

1、10 所人

语文、数学、英

息化负责人；

（每校 2—3 人

2、兴宁市

3、教师发

兴宁市教师发展中心

关于寒假期间推广使用 AI 助学工具的通知

各中小学校：

寒假既是学生享受中国传统节日、欢度春节的美好时光，也是培养学生对知识探索兴趣、体验新质生产力发展成果的重要契机。为深入贯彻教育数字化战略，推动人工智能技术与教育教学深度融合，帮助学生体验 AI 赋能的新型学习方式，现就寒假期间推广使用 AI 助学工具有关事项通知如下：

一、工作目标

引导学生在假期合理使用 AI 助学工具，培养创新思维和信息素养， 探索高效学习新模式，提升自主学习能力。

二、工作要求

（一）加强宣传引导。各校应在放假前通过家长会、班级群等渠道，向学生和家长介绍 AI 助学工具的功能特点及使用方法，做好宣传推广工作。

（二）坚持自愿原则。学生使用 AI 助学工具应遵循自愿原则，不得强制要求，充分尊重学生和家长的选择权。

（三）注重使用规范。引导学生合理安排使用时间，注意用眼卫生，培养良好的数字化学习习惯，建议学生需家长监督下使用。

三、 其他事项

具体推进办法及相关工具介绍详见附件，请各校(特别是试点校)结合实际情况认真组织实施。

佛山市南海区教育发展研究院
佛山市南海区教师发展中心

关于召开暨南大学广东智慧教育研究院“人工智能赋能基础教育”合作项目启动仪式暨专题培训活动的通知

各镇（街道）教育发展研究中心、相关学校：

为深入落实《魅星领航 智学南海——南海区推进中小

学人工智能教育创新

要求，充分借助高校

能赋能基础教育项目

广东智慧教育研究院

动仪式暨专题培训活

一、会议时间

2025 年 12 月 23

二、会议地点

南海区南海中心

三、会议内容

1.领导致词；

2.区域合作：暨

暨南大学广东智慧教育研究院
深圳市教育信息技术中心
合作协议

江海区景贤实验学校首次用AI大模型覆盖九年级寒假作业全流程

AI作业“千人千面”激发学生学习热情

当教育数字化转型不断推进，“因材施教”这一质朴的教育理想正借助人工智能技术得以更好地实现。在此背景下，江海区景贤实验学校联合暨南大学广东智慧教育研究院等合作单位，依托共建的人工智能应用示范基地，在今年寒假首次将校企共同研发试用的AI大模型“九章爱学教学智能体”深度融入九年级各科作业全流程，开启了一场特别的教育实验。这场实验不设强制要求、没有教师线上督促，却在短短假期中收获了令人惊喜的数据，也引发了新的教育思考。

文/吕中言

完全自愿 45%的学生自发登录系统

“过去的寒假作业每位同学做的内容都差不多，难免会导致会的题浪费一些时间，不会的题还是不会。”景贤实验学校九年级学生阿俊说出了许多学生的心声。但这个寒假，他和同学们的体验完全不同——通过平板电脑或手机登录AI系统，每个人都能获得量身定制的作业任务。

这套由学校与企业共同研发的AI大模型，实现了两大核心功能：一是学生可以自主点播想要练习的知识点，二是系统能根据学校整体教学进度、教师授课特点以及学生个体学情，自动生成个性化作业。客观题和主观题先由AI初评再由教师进行复核指导，真正扮演了“教学助手”的角色。

值得一提的是，在完全自愿、不强制的假期环境下，九年级参与的学生中，45%的学生自发登录系统，累计完成2140次题目练习与答疑互动。阿俊同学更是以寒假狂刷87题的成绩登顶“自律之星”榜首。“系统推送的练习刚好是我的薄弱点，刷题不再盲目，反而有种闯关的快感。”阿俊说。

通过对练习数据的词频分析，一幅清晰的“学生自主查漏补缺图”浮现出来：化学科目中，“氢氧化钠的简单物理性质及用途”被练习56次，“浓硫酸的吸水性”和“中考化学攻坚的重点。英语次）、“discussion”（目中，“三角函数值”角”（32次）等拉分对象。

“千人千面” AI让每个孩子都有自己的学习节奏

“传统作业，尤其是寒暑假这种周期长，总量相对较大的作业，往往会存在‘一刀切’的情况，一定程度上导致学习情况好的学生吃不饱，学习情况一般的学生跟不上。但在AI技术的帮助下，能让假期作业高效实现‘千人千面’。”景贤实验学校校长戴旌认为，在学校统筹下，AI技术和移动设备的介入，能有效让学生的学习从被动完成任务转向主动查漏补缺。

景贤实验学校九年级学生阿滔就是典型的“偏科突破者”——寒假期间单化学一科就刷了61次。“以前总觉得化学方程式记不住，现在AI把知识点布置成针对性的作业，反复推演，慢慢就开窍了。”他说。

如果说AI助力学科学习尚在意料之中，那么后台数据的另一个发现则堪称意外之喜。在紧张的初三备考阶段，原本以为学生只会聚焦中考科目，但系统显示，传统文化类知识点备受青睐：“楠书与行书”被练习45次，“隶书与草书”被练习36次，“甲骨文与篆书”被练习35次。“这说明学生没有因为中考压力变得麻木。”戴旌表示，AI不仅是提分工具，更应该是拓宽眼界的窗口。

AI寒假作业也带来了新的教育思考。戴旌指出，本次实验一定程度上证明了AI可以作为常规工具，与物理世界同频共振，持续在多种教学场景下进行应用，而不是简单的低频次体验，当AI分担了批改、诊断、推送等教学任务，教师的角色也能得以重新定位：教师是“核心”，AI是“助手”——前者腾出更多的精力与时间进行设计、引导、陪伴与价值塑造，后者则负责精准诊断与个性供给。例如，主观题先由AI批改，教师再进行精细化和更多维度的复核，既发挥效率优势，又保留人文关怀。“AI可以成为强大的教学助手，但教育的温度、真诚的对话、价值的引领，终究要靠人与人的相遇来完成。”戴旌说。

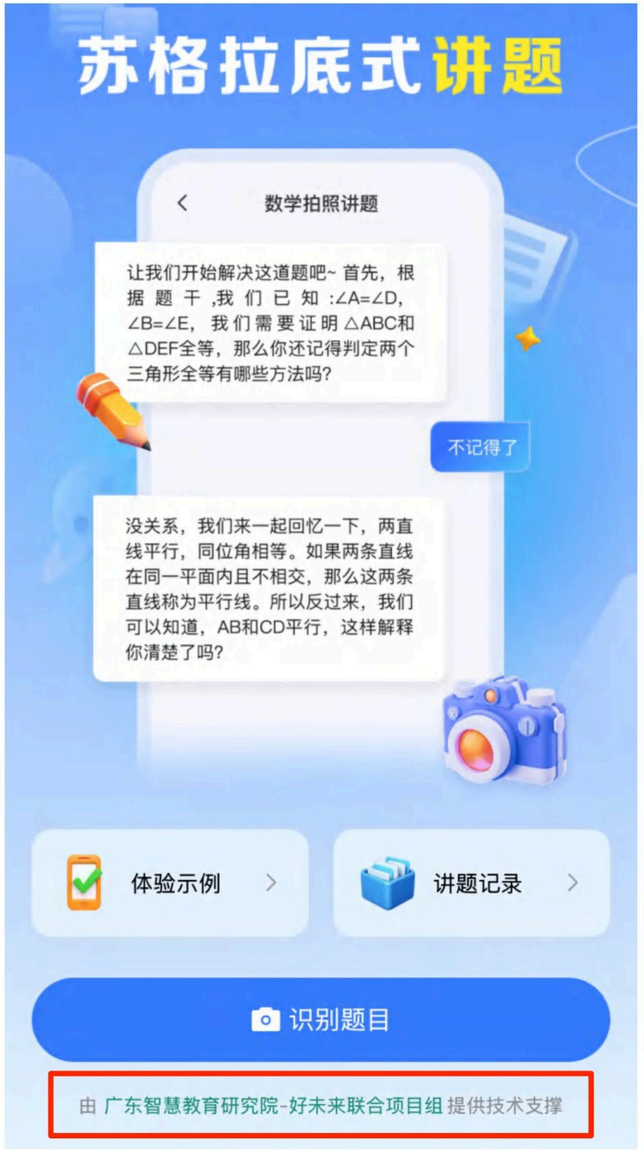
传统假期，家长业、查进度是常态。开始向“助学者”转到他在书桌前坐着，哪些知识点反复出现从‘数据消费者’转学生训练成‘刷题机生的时间，激发学生能看见更广阔的世女士说。

Application: National Smart Education Platform for K-12

https://basic.smartedu.cn/AIEducation

Application: Guangdong Smart Education Platform for K-12

Application: Guangdong Digital Textbook Platform



Application: Shenzhen Education Public Service Platform

Application: Guangxi Smart Education Platform

应用情况说明

（应用情况说明无格式要求，此表只供参考）

应用成果名称	中小学智慧学习服务平台关键技术及应用		
应用单位名称	广西东信数教科技有限公司		
应用单位联系人	万士桥	联系电话	18107710058
成果应用起始时间	2025年4月		
应用情况	成果的具体应用情况以及取得经济效益和社会（环境、生态）效益情况： 好未来“九章爱学”应用作为广西智慧教育平台（桂教通）的重要组成部分，以自主研发的多模态智能学伴能力为核心引擎，通过深度对接“桂教通”平台，构建起覆盖教、学、管的全场景服务体系。 “九章爱学”创新性地将大模型能力与广西自治区级教育平台深度融合，聚焦数学学科智能化服务，上线以来为广西全区小、初、高中学校的在校学生提供7×24小时精准答疑服务，基于学生作答数据构建个性化学习路径库，实现“千人千面”的精准练习推荐，开创了大规模个性化教育服务新模式。		
声明	我单位保证上述提供的应用情况真实无误。如有不符，本单位愿意承担相关责任并接受相应的处理。 单位财务专用章：　　法人单位公章： 年　月　日　　年　月　日		

注：如表中所填内容不涉及经济效益情况，只需加盖应用单位法人公章。

02

Multimodal Teaching Resource Generation with Educational Agents

Problem: Core Bottlenecks Facing AI Tutors

Resource supply dilemma

High-quality resources depend on manual creation — costly, slow and hard to scale.

Rigid resource formats

Static documents and finished videos are hard to decompose, recombine and adapt at fine granularity.

Limited generation capability

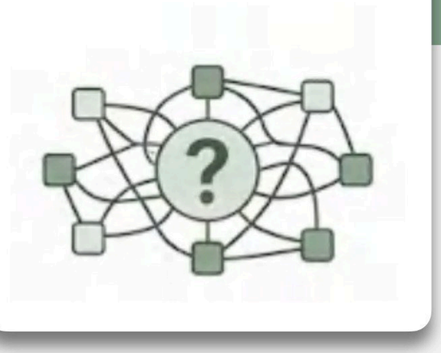
Most existing techniques stitch together material fragments, lacking pedagogical planning.

“Teaching” and “learning” disconnected

Resource generation is not systematically coupled with learner profiles (knowledge tracing / cognitive diagnosis).

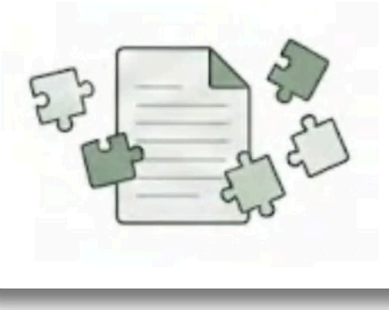
Key Technical Challenges

Collaborative framework



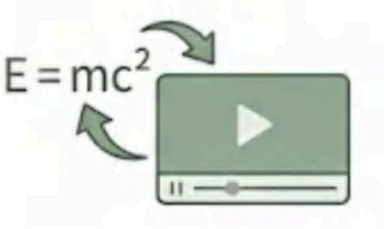
How to couple learner profiles with generation decisions?

Multimodal lesson plans



How to represent lesson plans structurally and support class heterogeneity?

STEM explainer videos



How to precisely align symbolic reasoning with visual presentation?

Poetry teaching videos

How to balance pedagogical clarity and aesthetic experience?

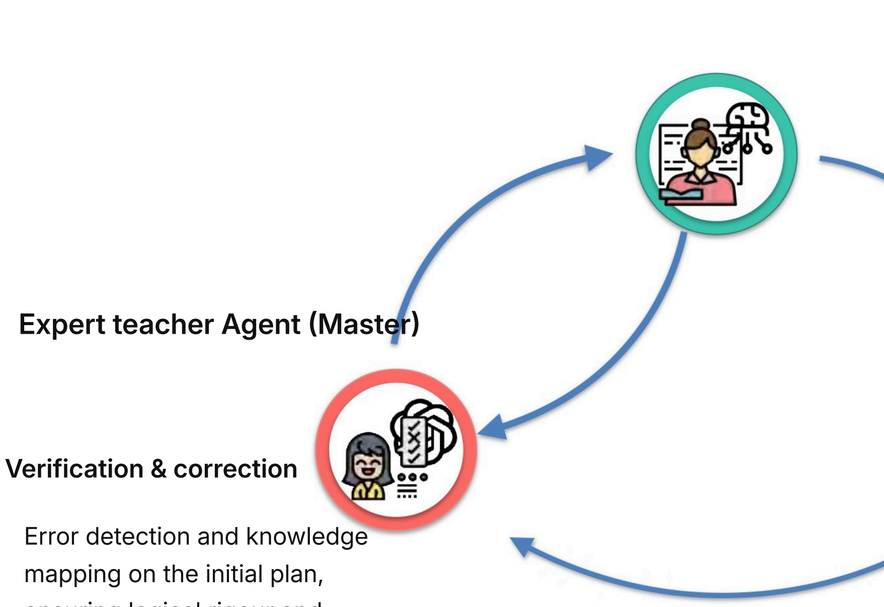
Research: Knowledge-Enhanced, Multi-Agent Lesson Plan Generation

Based on multi-agent collaboration, we build a high-quality lesson-plan generation framework and design a novice–expert collaboration and iteration mechanism, fixing loose structure and pedagogical inaccuracy.

Example of an existing lesson-plan task

- Over-reliance on manual work: no automated evaluation or feedback loop — plans need heavy manual refinement.
- Insufficient knowledge depth: no authoritative textbooks or curriculum standards — limited expertise.
- Poor structural consistency: generated plans jump in logic and rarely meet real classroom standards.

Multi-role collaborative COMA framework



Novice teacher Agent

Planning & finalization

Fine-tuned model (Qwen2-72B); decomposes subtasks and builds the initial lesson-plan framework.

Expert teacher Agent (Master)

Verification & correction

Error detection and knowledge mapping on the initial plan, ensuring logical rigour and pedagogical fit.

Senior teacher Agent (Veteran)

Knowledge injection

Knowledge base built from 574 latest textbooks; extracts core knowledge points and teaching strategies.

Xiaoli Zeng, Ying Zheng, Shuyan Huang, Zitao Liu, Mi Tian, Mingliang Hou, Jiaqi Zheng, Wenzhou Dou. “COMA: A Collaborative Multi-Role Agent Framework for Automated Lesson Plan Generation”. WWW 2026.

Research: Knowledge-Enhanced, Multi-Agent Lesson Plan Generation

Based on a multi-dimensional evaluation system covering five subjects, we systematically verify the generation quality and pedagogical fit of the multi-role architecture in complex teaching scenarios.

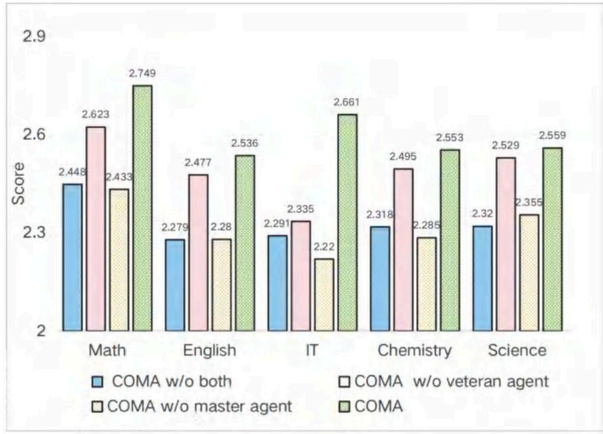
Cross-subject generation quality evaluation

- Evaluation scope: five core subjects — maths, English, chemistry, physics, science and IT
- Baselines: GPT-4o, Claude 3.5, Self-Critique and other mainstream methods
- Metrics: expert-designed, multi-dimensional indicators

Model	Math	English	IT	Chem	Science
GPT-4o★	2.481 κ=0.80	2.154 κ=0.68	2.273 κ=0.73	2.344 κ=0.78	2.399 κ=0.90
Claude3.5★	2.204 κ=0.83	1.985 κ=0.77	2.032 κ=0.83	2.224 κ=0.67	2.289 κ=0.74
GLM-4★	2.403 κ=0.80	2.149 κ=0.81	2.154 κ=0.70	2.303 κ=0.66	2.371 κ=0.80
Qwen2-72B★	2.166 κ=0.68	1.831 κ=0.89	2.012 κ=0.90	2.197 κ=0.70	2.132 κ=0.93
Self-Critique*	2.514 κ=0.74	2.167 κ=0.70	2.204 κ=0.86	2.355 κ=0.66	2.322 κ=0.79
Prompt4ChatGPT*	2.219 κ=0.84	1.991 κ=0.85	2.061 κ=0.89	2.192 κ=0.85	2.063 κ=0.77
COMA♣	**2.749** κ=0.77	**2.536** κ=0.71	**2.661** κ=0.88	**2.553** κ=0.67	**2.599** κ=0.86

Core-mechanism ablation and expert blind test

- Full removal (w/o both): internal knowledge alone cannot produce an acceptable plan; scores fall sharply across subjects.
- w/o Veteran: without textbook context, the model misses structural problems such as going off-topic.
- w/o Master: no iterative correction — hallucinations and logical gaps remain.



Module	Win (%)	Loss (%)	Tie (%)	Kappa
Textbook Analysis	40.00%	26.67%	33.33%	0.792
Lesson Objectives	23.53%	18.63%	57.84%	0.864
Key & Difficult Point	28.43%	17.65%	53.92%	0.676
Materials	40.78%	5.83%	53.40%	0.641
Lesson Procedure	27.45%	23.53%	49.02%	0.658

- Expert blind test: a double-blind comparison against lesson plans written by frontline teachers (Win/Tie rate) — over 80% of plans reach or exceed human-expert level.

Xiaoli Zeng, Ying Zheng, Shuyan Huang, Zitao Liu, Mi Tian, Mingliang Hou, Jiaqi Zheng, Wenzhou Dou. “COMA: A Collaborative Multi-Role Agent Framework for Automated Lesson Plan Generation”. WWW 2026.

A Library of 42 Frequently Used Teaching Models

适用学科

语文 | 数学 | 英语 | 物理 | 化学 | 生物 | 科学 | 信息科技

适用小初高学段

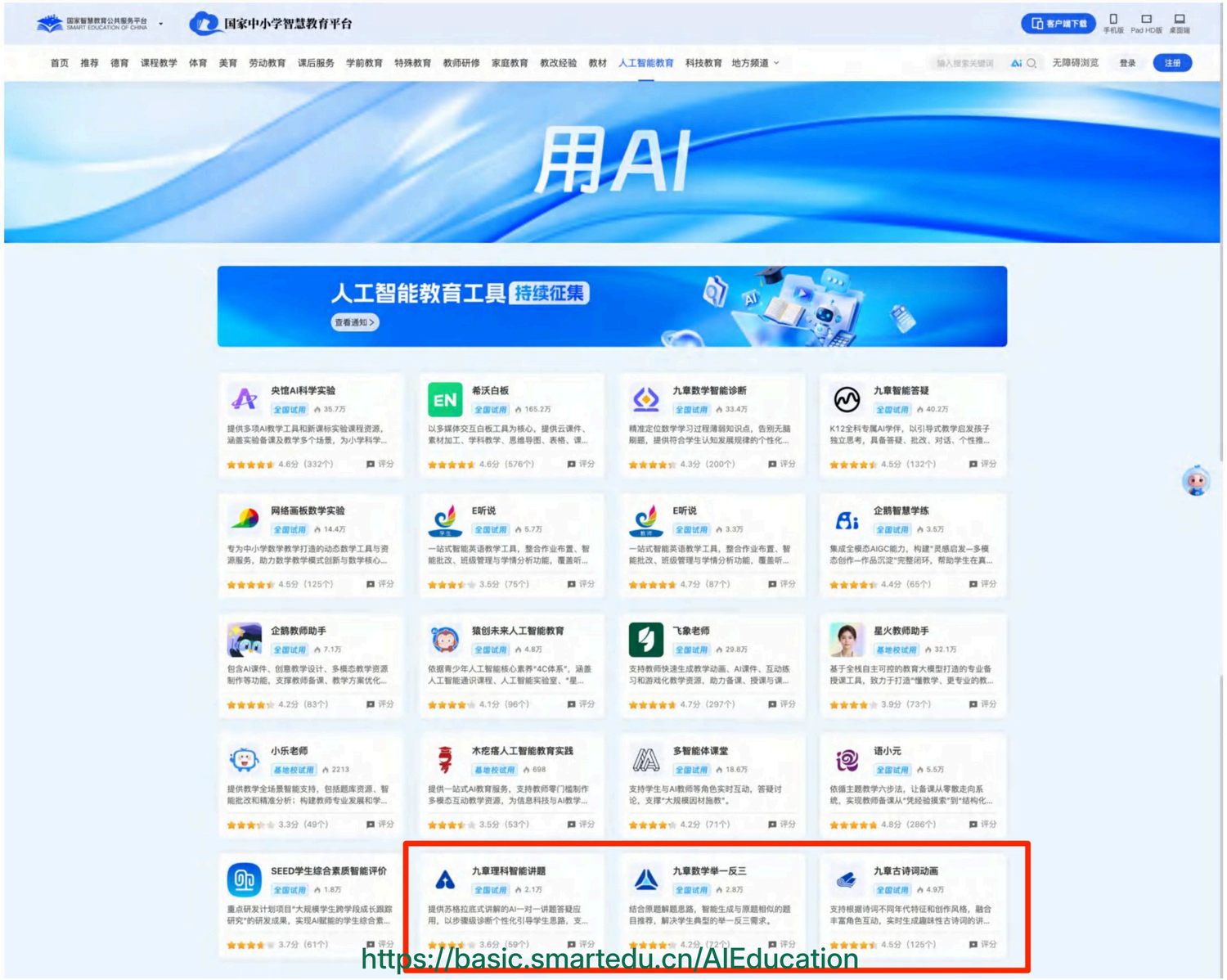
讲授式教学	比较式教学	自学——辅导教	以文献为导向的自我学习	项目式学习	基于问题的教学
探究式教学	案例式教学	任务驱动式教学	语文阅读课教学模式	情景式教学	
协作式教学	讲座式教学	自定义	角色扮演教学模式	基于资源支持的英语听说教学模式	
支架式教学	辩论式教学	启发式教学教学	精加略教学模式	以言语交际为中心的新授课教学模式	
单点式教学	“设问—导学”教学模式		古诗阅读教学模式	多媒体网络下英语“任务型”教学模式	
换位式教学	“问题－实验－探究”教学模式		作文教学模式	以言语交际为中心的复习课教学模式	
头脑风暴式教学	激励——导学教学模式		概念获得式教学	识、读、写三结合教学模式	
研究型教学	以言语交际为中心的巩固练习课教学模式		抛锚式教学	运用几何画板讲授数学公理、定理	

适用小学学段

图文归纳模式 | “随文识字”教学模式 | 低年级识字教学模式 | 小学语文读写教学模式 | 小学英语拓展听说教学模式

Yuejiao Xiangyun Launch — Serving All K-12 Teachers in Guangdong

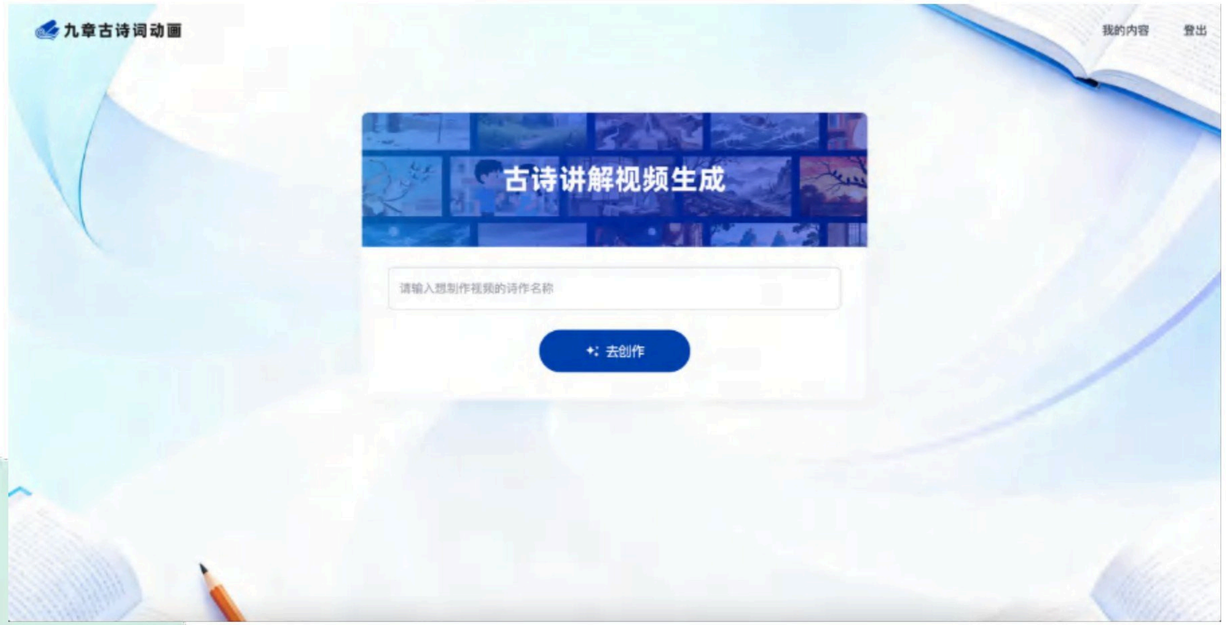
Application: National Smart Education Platform



https://basic.smartedu.cn/AIEducation

National Platform — Jiuzhang Classical Poetry Animation Agent

Generative AI synthesizes explainer animation videos for classical poems in real time. The videos feature rich character interaction and generate highly watchable, engaging animation that reflects the era and style of each poem — building a strong interest in learning classical poetry.



National Platform — Jiuzhang Similar-Question Agent

Leveraging the math item analysis of a domain-specific large model, each item is described with multi-dimensional tags. Based on the original item, similar items are recommended — meeting students' typical need to generalize from one example and supporting targeted practice.

03

Precision Learning Companion (OpenClaw) for Students

A Large Model Is Not an Agent

WHAT MOST PEOPLE THINK AI IS

Ask it, it answers

Input a question → output an answer

One question, one answer — purely passive

No memory, no agency

Like an old phone that can only make calls

VS

A REAL AGENT

It thinks proactively and acts on its own

Perceives the environment → makes a plan

Calls tools → executes continuously

Has memory, iterates, reaches the goal

Like a smartphone that can do countless things

> Treating AI as a "question-answering machine" is like using a smartphone as an old phone that can only make calls — you miss everything it can really do.

M Memory — keeps context across sessions

A Autonomy — plans and acts

T Tool use — calls external capabilities

Large model (LLM) + memory + action + tools = a real agent

From “AI Assists People” to “the Agent Just Does It”

Research 2023–2025 vs. the new paradigm of 2026+

2023 → 2024 → 2025 → 2026 · the year of agents → …

2023 — 2025

AI assists people

Prompt engineering · Human–AI collaboration · Human-led

Research focuses on how to write prompts so that AI helps you finish tasks. In essence, humans learned to use AI — AI is an assisting tool.

You tell the AI how to do the job

2026+ — THE AGENT ERA

The agent just does it

Goal setting · Autonomous execution · Human authorization

You set the goal; the agent plans and executes by itself. In essence, humans authorize and AI delivers — no prompt engineering needed.

You tell the AI what you want — it works out the rest

> From “learning to drive” to “calling a ride” — you used to learn to drive (prompting); now you just say where you want to go, and the car drives itself.

From “human + AI tool” to “human + agent” · assist → delegate · collaborate → authorize

From the Platform Economy to the Agent Economy

A paradigm shift in internet business models

PLATFORM ECONOMY

The platform economy

- Super-app monopoly: Taobao, Didi, Meituan and Douyin aggregate transactions.
- Data monopoly: platforms capture consumer data and monetize advertising; users gain nothing.
- Bidding costs: merchants keep bidding for ads while margins are squeezed.

Centralized — the platform is the only middleman

VS

AGENT ECONOMY

The agent economy

- Agent delegation: people no longer use apps directly — agents access services via APIs.
- Decentralization: individuals transact directly with merchant agents; no platform needed.
- Trust comes first: telling which merchant agent is trustworthy becomes the core issue.
- Value re-anchored: new value must be found in the new ecosystem.

Decentralized — the agent is the new connector

> From “platform commission” to “trust premium” — the most valuable thing is no longer traffic, but the ability to judge which merchant agents deserve trust.

Xiao Jing Long — Your Personal AI Study Companion

Not just an AI tool — a companion that understands how you learn

TRADITIONAL AI TOOLS

Traditional study tools

- Q&A only — students ask, AI responds passively
- Use and leave — no history, every session starts from zero
- One size fits all — the same answer for every student
- Cold tool — answers only, ignoring emotion and state

Tool mindset — search-engine style Q&A

VS

XIAO JING LONG

Xiao Jing Long study companion

- Proactive understanding — tracks knowledge gaps, cognitive bias and reasoning paths
- Memory accumulation — mistakes, preferences and materials become private learning assets
- Warm company — adapts to your personality, pace and emotional state
- Dedicated evolution — free Skill configuration, deeply customized for you

Companion mindset — a private AI partner that understands you

> From “tool” to “partner” — what Xiao Jing Long sees is not an answer, but the knowledge gap and cognitive bias behind the question, and the part of you that should be lit up next.

Traditional AI: Question → Answer **Xiao Jing Long: Understand → Guide → Grow**

Four Core Capabilities

How Xiao Jing Long redefines the “AI study companion”

B Brain Mapping

Across every question, every mistake, every hesitation and every step forward, it keeps understanding your knowledge structure, ability boundaries and thinking style. What it sees is not the answer, but the knowledge gap and cognitive bias behind the question.

A deep grasp of what learning really is

I Private Assets

What you said, the questions you solved, the mistakes you made, the preferences you formed, the materials you uploaded, the traces of your revision — all of it accumulates and settles inside Xiao Jing Long, and becomes your most valuable private learning wealth.

Compounding your own growth marks

C Learning Companion

When you don't understand, it explains it clearly; when you get it wrong, it points out the problem; when you're anxious, it steadies the pace; when you want to give up, it pulls you back. It adjusts the way it walks with you to your personality, your pace and your state.

A warm fellow traveller at your pace

P Personal Companion

Not just an AI tool, but a one-of-a-kind private partner, different for every person. Skills are freely configurable and deeply customizable — your own exclusive learning companion: break the constraints, explore without limits.

Skills freely configured, exclusivity that evolves

The essence of the four capabilities — from “giving you the answer” to “understanding how you grow”: brain mapping is perception, private assets are memory, learning companion is empathy, personal companion is evolution. That is what real Agent thinking looks like.

New Features & Highlights

How Xiao Jing Long makes the learning experience feel brand new

Three new features

S

My Space

A personal creative space
Create and organise freely

E

Error Notebook

Organise and learn from mistakes
Target your weak spots precisely

R

Companion Roles

Choose the companion that warms you
Switch styles whenever you like

Four core highlights

01

Break the constraints, explore without limits

Summon it whenever you need it, so learning is no longer bound by the format of a system.

02

A thousand learners, a thousand companions

Skills are freely configurable and deeply customizable — your own private learning companion.

03

No more question-swamp, precise improvement

Refuse low-efficiency drilling and give your time to the core knowledge points that really need work.

04

Zero-friction setup, start instantly

No learning curve at all — one tap and you're in, starting smart learning effortlessly.

> Xiao Jing Long's core proposition — not to make you grind harder, but to let AI understand you more intelligently. Every single interaction is a growth path tailored to you.

小精龙 AI学伴
私有财产，沉淀专属成长印记
大脑映射，深刻理解学习本质
学习伴侣，随你节奏的温暖同行者
不只工具，更是私人AI学伴
平板扫码体验
iPad版下载
安卓HD版下载

04

Summary and Outlook

Three Key Judgments

In the AI era of education, every teacher needs to see these trends clearly

01

AI is an amplifier, not a replacement

AI will not replace teachers — but teachers who use AI will replace those who don't.

AI is a teaching amplifier: it copies an excellent teacher's ability a thousand times over and makes classroom efficiency leap. But the core of education — guidance, encouragement, care — will always have to be done by a person.

Teachers are irreplaceable, but the toolchain must be upgraded

02

The essence stays, the method is rebuilt

The essence of education has never changed — teach according to aptitude, spark potential, ignite curiosity.

What changes is the means: from chalk and blackboard to electronic whiteboards, from one-size-fits-all lesson plans to AI-powered personalised recommendation. The unchanging ideal is, with technology's help, truly landing on the ground.

The ideal of teaching to aptitude will be lit by technology

03

The next three years are the watershed

The choice you make now determines where you stand three years from now.

Three years ago, online teaching was a passive adaptation. Three years from now, AI-assisted teaching is proactive evolution. Teachers who embrace AI today are becoming the first cohort of beneficiaries of this transformation.

The window is closing — act now, no time to lose

The underlying logic of all three judgments — AI is not here to replace you, but to amplify your teaching ability a hundredfold. The question is: are you ready to press the “amplify” button?

A Real Example

Re: Nomination for Lifetime Achievement Award (回复：终身成就奖提名) 收件箱

总结此邮件

Bruce McLaren 6月5日
Thank you, Didith. I have added Tanja to our ball...

Maria Mercedes (Didith) T. Rodrigo 6月6日
Hi Bruce. I'm so sorry, I didn't realize I had to as...

Maria Mercedes (Didith) T. Rodrigo 6月6日
Hi Bruce. Tanja wrote back. She would NOT like t...

Kalina Yacef 6月6日
Thanks Didith for writing the nomination text for...

Bruce McLaren 6月6日
Thanks, Kalina. I definitely would be in favor of g...

Bruce McLaren 6月6日
One other reply to your excellent comments, Kali...

Akihiro Kashihara 6月7日
Dear Bruce, Sorry for my late reply. I fully agree...

Kalina Yacef 6月7日

回复所有人 转发

Bruce McLaren 6月15日
发给 Erin, Kalina, me, Maria, akihir...

正在翻译...

Hi Everyone —

I have been trying to get the updated email list from Seiji but he has been unresponsive. It is getting awfully late to conduct the vote. It is already too late, I think, to solcit any more nominees, given the effort that takes.

Could someone on the committee volunteer to compile the email list of current EC members (+ Journal Editors + President) and their email addresses? I would really like to send out the ballot I created by tomorrow, at the latest, but I am traveling, have meetings, and can't do it now.

Thanks,

... bruce ...

Carnegie Mellon University

Human-Computer Interaction Institute

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Home > People > Faculty > Bruce M. McLaren

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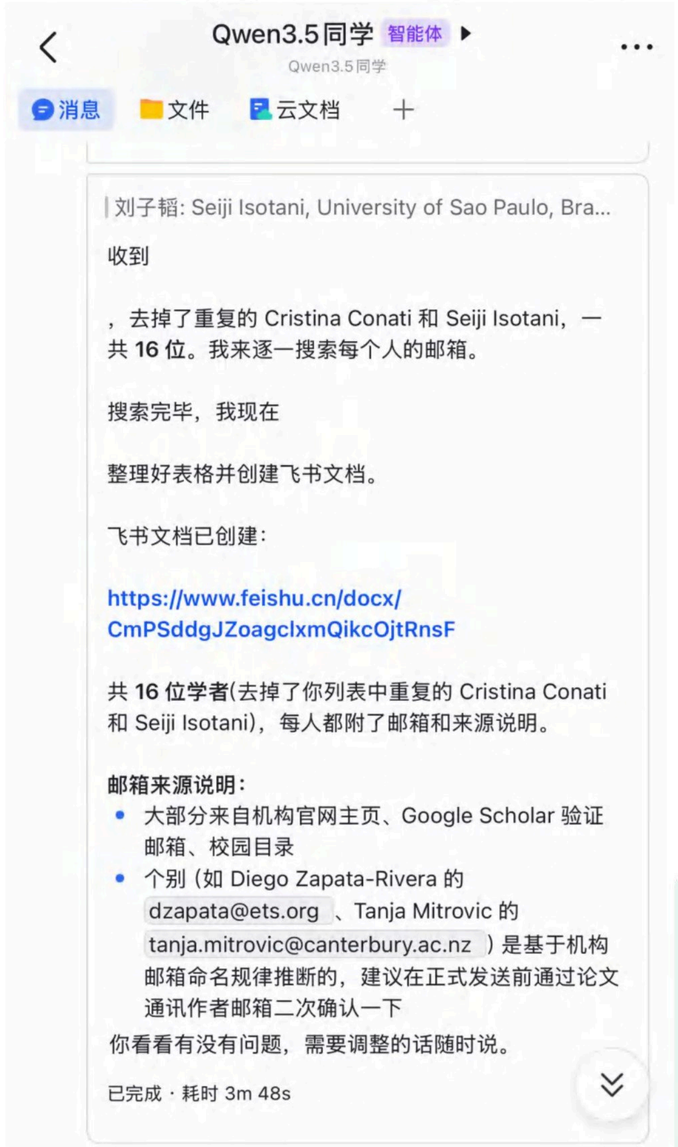
Links
- Personal Website
- McLearn Lab Website
- Google Scholar

Research Areas
- Learning Sciences and Educational Technologies
- Human-Centered AI
- Responsible Computing, Ethics and Policy
- Games and Play

A Real Example



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Thank You!

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